

RW3 Module Report - sheet_cme

INTERPRETATION GUIDE

- ADF $p > 0.05$ and KPSS $p < 0.05$ -> likely $I(1)$ -like; prefer differences/cointegration models.
- ADF $p < 0.05$ and KPSS $p > 0.05$ -> stationary; level models are more admissible.
- Ljung-Box $p < 0.05$ -> autocorrelation; add lag structure.
- BP/White $p < 0.05$ -> heteroskedasticity; use robust or HAC standard errors.
- JB $p < 0.05$ -> non-normality; use robust inference and avoid strict normality assumptions.
- Stability flag=1 -> potential breaks/drift; use rolling or split-sample robustness.
- For retail transmission, compare no-promo vs promo-controlled estimates.

RUN NOTES

module=sheet_cme

xlsx_files=1

png_files=3

Interpretation: read test diagnostics first, then coefficients/effects with robustness context.

Table: sheet_cme_output_raw

Date	CME III UAH
2026-01-27	633.94191
2026-01-26	635.007552
2026-01-23	634.1188229999999
2026-01-22	635.981007
2026-01-21	637.5374280000001
2026-01-20	638.20005
2026-01-16	639.608398
2026-01-15	635.641864
2026-01-14	637.712474
2026-01-13	638.879304
2026-01-12	637.52036
2026-01-09	638.837344
2026-01-08	636.033795
2026-01-07	641.856072
2026-01-06	640.129872
2026-01-05	635.6818259999999
2026-01-02	635.5034069999999
2025-12-31	645.142316
2025-12-30	670.420252
2025-12-29	667.5509189999999
2025-12-26	665.058966
2025-12-24	663.9217309999999
2025-12-23	665.098596
2025-12-22	666.252537
2025-12-19	667.660798
2025-12-18	668.523336
2025-12-17	667.785384
2025-12-16	669.162384
2025-12-15	667.080816
2025-12-12	671.7036690000001
2025-12-11	671.0026439999999
2025-12-10	673.253448
2025-12-09	671.8579000000001
2025-12-08	670.383798
2025-12-05	675.7844759999999
2025-12-04	679.4747400000001
2025-12-03	668.8803600000001
2025-12-02	726.5767079999999
2025-12-01	725.2845599999999
2025-11-28	725.71616

Table: sheet_cme_output_clean

source	date	raw_product	product	standardized_type	brand	region	price	unit_ok
CME	2026-01-27	CME Class III	Молоко питне	milk	nan	US	633.94191	1
CME	2026-01-26	CME Class III	Молоко питне	milk	nan	US	635.007552	1
CME	2026-01-23	CME Class III	Молоко питне	milk	nan	US	634.1188229999999	1
CME	2026-01-22	CME Class III	Молоко питне	milk	nan	US	635.981007	1
CME	2026-01-21	CME Class III	Молоко питне	milk	nan	US	637.5374280000001	1
CME	2026-01-20	CME Class III	Молоко питне	milk	nan	US	638.20005	1
CME	2026-01-16	CME Class III	Молоко питне	milk	nan	US	639.608398	1
CME	2026-01-15	CME Class III	Молоко питне	milk	nan	US	635.641864	1
CME	2026-01-14	CME Class III	Молоко питне	milk	nan	US	637.712474	1
CME	2026-01-13	CME Class III	Молоко питне	milk	nan	US	638.879304	1
CME	2026-01-12	CME Class III	Молоко питне	milk	nan	US	637.52036	1
CME	2026-01-09	CME Class III	Молоко питне	milk	nan	US	638.837344	1
CME	2026-01-08	CME Class III	Молоко питне	milk	nan	US	636.033795	1
CME	2026-01-07	CME Class III	Молоко питне	milk	nan	US	641.856072	1
CME	2026-01-06	CME Class III	Молоко питне	milk	nan	US	640.129872	1
CME	2026-01-05	CME Class III	Молоко питне	milk	nan	US	635.6818259999999	1
CME	2026-01-02	CME Class III	Молоко питне	milk	nan	US	635.5034069999999	1
CME	2025-12-31	CME Class III	Молоко питне	milk	nan	US	645.142316	1
CME	2025-12-30	CME Class III	Молоко питне	milk	nan	US	670.420252	1
CME	2025-12-29	CME Class III	Молоко питне	milk	nan	US	667.5509189999999	1
CME	2025-12-26	CME Class III	Молоко питне	milk	nan	US	665.058966	1
CME	2025-12-24	CME Class III	Молоко питне	milk	nan	US	663.9217309999999	1
CME	2025-12-23	CME Class III	Молоко питне	milk	nan	US	665.098596	1
CME	2025-12-22	CME Class III	Молоко питне	milk	nan	US	666.252537	1
CME	2025-12-19	CME Class III	Молоко питне	milk	nan	US	667.660798	1
CME	2025-12-18	CME Class III	Молоко питне	milk	nan	US	668.523336	1
CME	2025-12-17	CME Class III	Молоко питне	milk	nan	US	667.785384	1
CME	2025-12-16	CME Class III	Молоко питне	milk	nan	US	669.162384	1
CME	2025-12-15	CME Class III	Молоко питне	milk	nan	US	667.080816	1
CME	2025-12-12	CME Class III	Молоко питне	milk	nan	US	671.7036690000001	1
CME	2025-12-11	CME Class III	Молоко питне	milk	nan	US	671.0026439999999	1
CME	2025-12-10	CME Class III	Молоко питне	milk	nan	US	673.253448	1
CME	2025-12-09	CME Class III	Молоко питне	milk	nan	US	671.8579000000001	1
CME	2025-12-08	CME Class III	Молоко питне	milk	nan	US	670.383798	1
CME	2025-12-05	CME Class III	Молоко питне	milk	nan	US	675.7844759999999	1
CME	2025-12-04	CME Class III	Молоко питне	milk	nan	US	679.4747400000001	1
CME	2025-12-03	CME Class III	Молоко питне	milk	nan	US	668.8803600000001	1
CME	2025-12-02	CME Class III	Молоко питне	milk	nan	US	726.5767079999999	1
CME	2025-12-01	CME Class III	Молоко питне	milk	nan	US	725.2845599999999	1
CME	2025-11-28	CME Class III	Молоко питне	milk	nan	US	725.71616	1

Table: sheet_cme_output_daily_variants

source	date	product	standardized_type	brand	region	unit_ok	price_real	price_linear	price_pchip	imputed_flag_linear	imputed_flag_pchip	
CME	2022-01-03	Молоко	питне	milk	nan	US	1	502.464444	nan	nan	0	0
CME	2022-01-04	Молоко	питне	milk	nan	US	1	502.464444	nan	nan	0	0
CME	2022-01-05	Молоко	питне	milk	nan	US	1	563.9684159999999	nan	nan	0	0
CME	2022-01-06	Молоко	питне	milk	nan	US	1	560.358591	nan	nan	0	0
CME	2022-01-07	Молоко	питне	milk	nan	US	1	557.063979	nan	nan	0	0
CME	2022-01-08	Молоко	питне	milk	nan	US	1	nan	nan	nan	0	0
CME	2022-01-09	Молоко	питне	milk	nan	US	1	nan	nan	nan	0	0
CME	2022-01-10	Молоко	питне	milk	nan	US	1	557.378379	nan	nan	0	0
CME	2022-01-11	Молоко	питне	milk	nan	US	1	560.170926	nan	nan	0	0
CME	2022-01-12	Молоко	питне	milk	nan	US	1	560.6395339999999	nan	nan	0	0
CME	2022-01-13	Молоко	питне	milk	nan	US	1	563.012336	nan	nan	0	0
CME	2022-01-14	Молоко	питне	milk	nan	US	1	562.5104160000001	nan	nan	0	0
CME	2022-01-15	Молоко	питне	milk	nan	US	1	nan	nan	nan	0	0
CME	2022-01-16	Молоко	питне	milk	nan	US	1	nan	nan	nan	0	0
CME	2022-01-17	Молоко	питне	milk	nan	US	1	nan	nan	nan	0	0
CME	2022-01-18	Молоко	питне	milk	nan	US	1	568.90806	nan	nan	0	0
CME	2022-01-19	Молоко	питне	milk	nan	US	1	574.736323	nan	nan	0	0
CME	2022-01-20	Молоко	питне	milk	nan	US	1	575.17695	nan	nan	0	0
CME	2022-01-21	Молоко	питне	milk	nan	US	1	574.685844	nan	nan	0	0
CME	2022-01-22	Молоко	питне	milk	nan	US	1	nan	nan	nan	0	0
CME	2022-01-23	Молоко	питне	milk	nan	US	1	nan	nan	nan	0	0
CME	2022-01-24	Молоко	питне	milk	nan	US	1	574.85743	nan	nan	0	0
CME	2022-01-25	Молоко	питне	milk	nan	US	1	575.442972	nan	nan	0	0
CME	2022-01-26	Молоко	питне	milk	nan	US	1	581.342424	nan	nan	0	0
CME	2022-01-27	Молоко	питне	milk	nan	US	1	586.335424	nan	nan	0	0
CME	2022-01-28	Молоко	питне	milk	nan	US	1	590.193644	nan	nan	0	0
CME	2022-01-29	Молоко	питне	milk	nan	US	1	nan	nan	nan	0	0
CME	2022-01-30	Молоко	питне	milk	nan	US	1	nan	nan	nan	0	0
CME	2022-01-31	Молоко	питне	milk	nan	US	1	586.328043	nan	nan	0	0
CME	2022-02-01	Молоко	питне	milk	nan	US	1	580.2076450000001	nan	nan	0	0
CME	2022-02-02	Молоко	питне	milk	nan	US	1	575.13858	nan	nan	0	0
CME	2022-02-03	Молоко	питне	milk	nan	US	1	580.5264199999999	nan	nan	0	0
CME	2022-02-04	Молоко	питне	milk	nan	US	1	585.19107	nan	nan	0	0
CME	2022-02-05	Молоко	питне	milk	nan	US	1	nan	nan	nan	0	0
CME	2022-02-06	Молоко	питне	milk	nan	US	1	nan	nan	nan	0	0
CME	2022-02-07	Молоко	питне	milk	nan	US	1	585.45344	nan	nan	0	0
CME	2022-02-08	Молоко	питне	milk	nan	US	1	579.315862	nan	nan	0	0
CME	2022-02-09	Молоко	питне	milk	nan	US	1	577.692854	nan	nan	0	0
CME	2022-02-10	Молоко	питне	milk	nan	US	1	580.2948299999999	nan	nan	0	0
CME	2022-02-11	Молоко	питне	milk	nan	US	1	579.8124799999999	nan	nan	0	0

Table: sheet_cme_output_series_long

source	date	product	standardized_type	brand	region	unit_ok	imputed_flag_linear	imputed_flag_pchip	series_variant	price	
CME	2022-01-03	Молоко	питне	milk	nan	US	1	0	0	real	502.464444
CME	2022-01-04	Молоко	питне	milk	nan	US	1	0	0	real	502.464444
CME	2022-01-05	Молоко	питне	milk	nan	US	1	0	0	real	563.9684159999999
CME	2022-01-06	Молоко	питне	milk	nan	US	1	0	0	real	560.358591
CME	2022-01-07	Молоко	питне	milk	nan	US	1	0	0	real	557.063979
CME	2022-01-10	Молоко	питне	milk	nan	US	1	0	0	real	557.378379
CME	2022-01-11	Молоко	питне	milk	nan	US	1	0	0	real	560.170926
CME	2022-01-12	Молоко	питне	milk	nan	US	1	0	0	real	560.6395339999999
CME	2022-01-13	Молоко	питне	milk	nan	US	1	0	0	real	563.012336
CME	2022-01-14	Молоко	питне	milk	nan	US	1	0	0	real	562.5104160000001
CME	2022-01-18	Молоко	питне	milk	nan	US	1	0	0	real	568.90806
CME	2022-01-19	Молоко	питне	milk	nan	US	1	0	0	real	574.736323
CME	2022-01-20	Молоко	питне	milk	nan	US	1	0	0	real	575.17695
CME	2022-01-21	Молоко	питне	milk	nan	US	1	0	0	real	574.685844
CME	2022-01-24	Молоко	питне	milk	nan	US	1	0	0	real	574.85743
CME	2022-01-25	Молоко	питне	milk	nan	US	1	0	0	real	575.442972
CME	2022-01-26	Молоко	питне	milk	nan	US	1	0	0	real	581.342424
CME	2022-01-27	Молоко	питне	milk	nan	US	1	0	0	real	586.335424
CME	2022-01-28	Молоко	питне	milk	nan	US	1	0	0	real	590.193644
CME	2022-01-31	Молоко	питне	milk	nan	US	1	0	0	real	586.328043
CME	2022-02-01	Молоко	питне	milk	nan	US	1	0	0	real	580.2076450000001
CME	2022-02-02	Молоко	питне	milk	nan	US	1	0	0	real	575.13858
CME	2022-02-03	Молоко	питне	milk	nan	US	1	0	0	real	580.5264199999999
CME	2022-02-04	Молоко	питне	milk	nan	US	1	0	0	real	585.19107
CME	2022-02-07	Молоко	питне	milk	nan	US	1	0	0	real	585.45344
CME	2022-02-08	Молоко	питне	milk	nan	US	1	0	0	real	579.315862
CME	2022-02-09	Молоко	питне	milk	nan	US	1	0	0	real	577.692854
CME	2022-02-10	Молоко	питне	milk	nan	US	1	0	0	real	580.2948299999999
CME	2022-02-11	Молоко	питне	milk	nan	US	1	0	0	real	579.8124799999999
CME	2022-02-14	Молоко	питне	milk	nan	US	1	0	0	real	583.680372
CME	2022-02-15	Молоко	питне	milk	nan	US	1	0	0	real	593.5384
CME	2022-02-16	Молоко	питне	milk	nan	US	1	0	0	real	586.0298
CME	2022-02-17	Молоко	питне	milk	nan	US	1	0	0	real	586.608984
CME	2022-02-18	Молоко	питне	milk	nan	US	1	0	0	real	591.47
CME	2022-02-22	Молоко	питне	milk	nan	US	1	0	0	real	595.0432940000001
CME	2022-02-23	Молоко	питне	milk	nan	US	1	0	0	real	605.553053
CME	2022-02-24	Молоко	питне	milk	nan	US	1	0	0	real	612.8901549999999
CME	2022-02-25	Молоко	питне	milk	nan	US	1	0	0	real	612.012508
CME	2022-02-28	Молоко	питне	milk	nan	US	1	0	0	real	612.012508
CME	2022-03-01	Молоко	питне	milk	nan	US	1	0	0	real	612.012508

Table: sheet_cme_output_descriptive_st

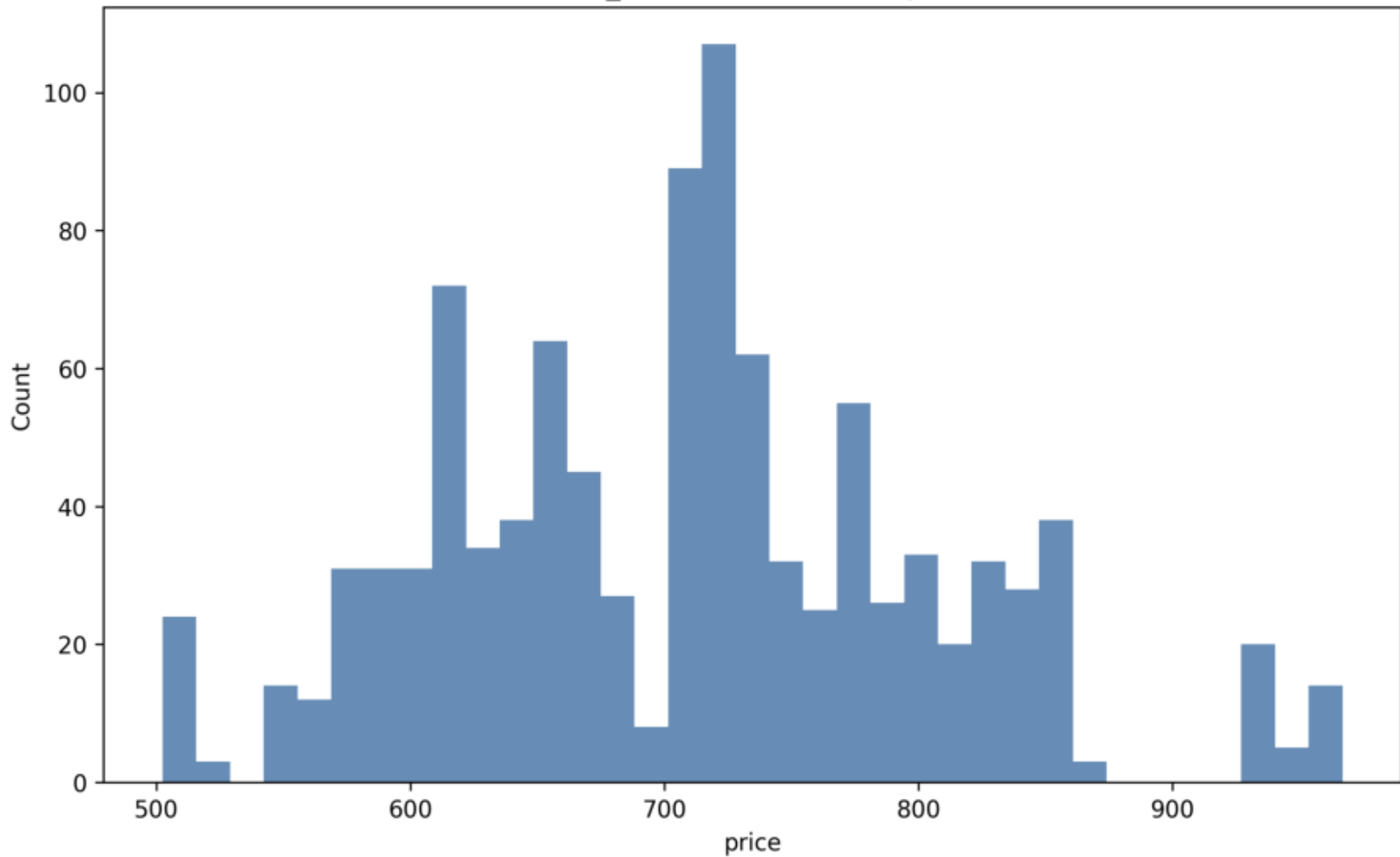
source	product	standardized_type	series_variant	count	missing	imputed_share		mean	median	std	min	max	q05	q25	q75	q95	skew	kurtosis	cv	rolling_vol_7	rolling_vol_30
CME	Молоко	питне	milk	linear	0	1486	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan
CME	Молоко	питне	milk	pchip	0	1486	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan	nan
CME	Молоко	питне	milk	real	1023	463	nan	709.2641239530792	713.331402	95.79149583199074	502.464444	966.9001849999999	564.4273536	635.788137	770.375485	854.4147727999999	0.2832414598568212	-0.07074089286782348	0.1350575795348247	0.01052681552669289	0.01516410362807883

Table: sheet_cme_output_tests

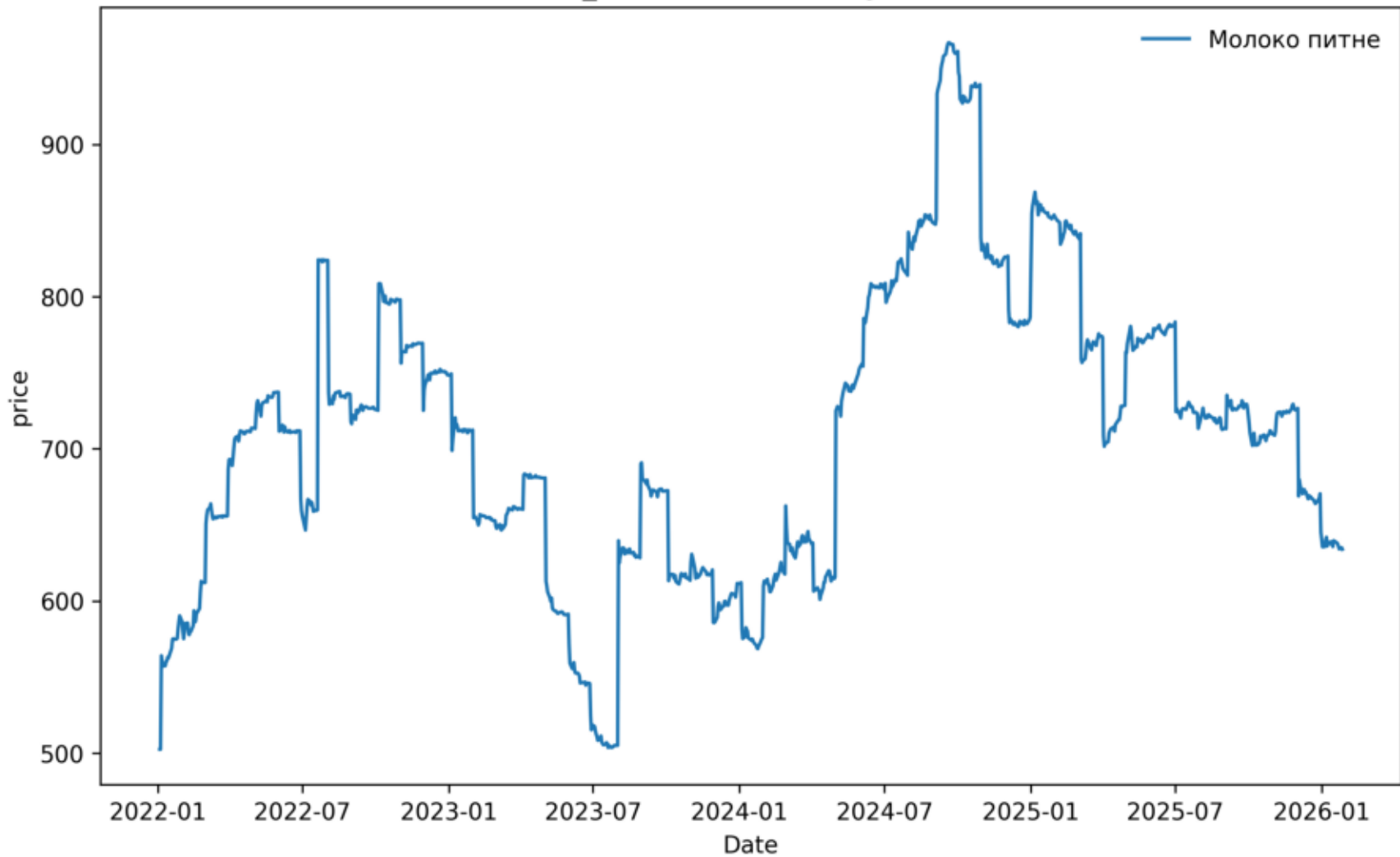
source	product	standardized_type	series_variant	n_obs	adf_p	kpss_p	ljungbox_p	bp_p	white_p	jb_p	stability_flag	stability_drift	action_label	integration_class	ac_risk	het_risk	non_normal_risk	stability_risk_class	recommended_action	recommended_model_family	avoid	diag_note
CME Молоко питне		milk	linear	0	nan	nan	nan	nan	nan	nan	0	nan	ok	ambiguous	low	low	low	low	Use differences or cointe...	Diagnostics-dependent	Unvalidated level OLS und...	Follow diagnostics-driven...
CME Молоко питне		milk	pchip	0	nan	nan	nan	nan	nan	nan	0	nan	ok	ambiguous	low	low	low	low	Use differences or cointe...	Diagnostics-dependent	Unvalidated level OLS und...	Follow diagnostics-driven...
CME Молоко питне		milk	real	214	0.198094907286723	0.04425555473715628	0.002331149919449167	0.5837986248407785	0.647495693005676	8.722972988440486e-233	1	91.95170941168215	non-stationary: differenc...	I(1)-like	high	low	high	high	Use differences or cointe...	Diagnostics-dependent	Unvalidated level OLS und...	Follow diagnostics-driven...

sheet_cme_distribution.png

sheet_cme: Distribution of price



sheet_cme: Time Series by Product



sheet_cme: Time Series by Standardized Type

